

```
import pandas as pd
```

```
df=pd.read_excel(r"/content/data.xlsx")
```

df

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola\nAlbania	AGO\nALB	45.985\n12.877	19.1\n57.2	Upper middle income\nUpper middle income
3	United Arab Emirates	ARE	11.044	88	High income
4	Argentina	ARG	17.716	59.9	High income
...	...	...	...	...	...
144	Samoa	WSM	26.172	15.3	Lower middle income
145	Yemen, Rep.	YEM	32.947	20	Lower middle income
146	South Africa\nCongo, Dem. Rep.	ZAF\nCOD	20.85\n42.394	46.5\n2.2	Upper middle income\nLow income
147	Zambia	ZMB	40.471	15.4	Lower middle income
148	Zimbabwe	ZWE	35.715	18.5	Low income

149 rows × 5 columns

```
id(df)
```

135150615602304

```
len(df)
```

149

```
df.shape
```

(149, 5)

```
df.head()
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola\nAlbania	AGO\nALB	45.985\n12.877	19.1\n57.2	Upper middle income\nUpper middle income
3	United Arab Emirates	ARE	11.044	88	High income
4	Argentina	ARG	17.716	59.9	High income

```
df.head(2)
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income

```
df.tail()
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
144	Samoa	WSM	26.172	15.3	Lower middle income
145	Yemen, Rep.	YEM	32.947	20	Lower middle income
146	South Africa\nCongo, Dem. Rep.	ZAF\nCOD	20.85\n42.394	46.5\n2.2	Upper middle income\nLow income
147	Zambia	ZMB	40.471	15.4	Lower middle income
148	Zimbabwe	ZWE	35.715	18.5	Low income

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 149 entries, 0 to 148
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   CountryName     149 non-null   object
1   CountryCode     149 non-null   object
2   BirthRate       149 non-null   object
3   InternetUsers   149 non-null   object
4   IncomeGroup     149 non-null   object
dtypes: object(5)
memory usage: 5.9+ KB
```

```
df.columns

Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
      'IncomeGroup'],
      dtype='object')
```

```
len(df.columns)

5
```

```
df.dtypes

0
CountryName  object
CountryCode  object
BirthRate    object
InternetUsers object
IncomeGroup  object

dtype: object
```

```
df['CountryName']
```

	CountryName
0	Aruba
1	Afghanistan
2	Angola\nAlbania
3	United Arab Emirates
4	Argentina
...	...
144	Samoa
145	Yemen, Rep.
146	South Africa\nCongo, Dem. Rep.
147	Zambia
148	Zimbabwe

149 rows × 1 columns

dtype: object

df[['BirthRate', 'InternetUsers']]

	BirthRate	InternetUsers
0	10.244	78.9
1	35.253	5.9
2	45.985\n12.877	19.1\n57.2
3	11.044	88
4	17.716	59.9
...	...	...
144	26.172	15.3
145	32.947	20
146	20.85\n42.394	46.5\n2.2
147	40.471	15.4
148	35.715	18.5

149 rows × 2 columns

df_numerical_data=df[['BirthRate', 'InternetUsers']]

df_numerical_data

	BirthRate	InternetUsers
0	10.244	78.9
1	35.253	5.9
2	45.985\n12.877	19.1\n57.2
3	11.044	88
4	17.716	59.9
...	...	...
144	26.172	15.3
145	32.947	20
146	20.85\n42.394	46.5\n2.2
147	40.471	15.4
148	35.715	18.5

149 rows × 2 columns

```
df_categorical_data = df[['CountryName', 'CountryCode', 'IncomeGroup']]
```

```
df_categorical_data.head()
```

	CountryName	CountryCode	IncomeGroup
0	Aruba	ABW	High income
1	Afghanistan	AFG	Low income
2	Angola\nAlbania	AGO\nALB	Upper middle income\nUpper middle income
3	United Arab Emirates	ARE	High income
4	Argentina	ARG	High income

```
df_numerical_data
```

	BirthRate	InternetUsers
0	10.244	78.9
1	35.253	5.9
2	45.985\n12.877	19.1\n57.2
3	11.044	88
4	17.716	59.9
...	...	...
144	26.172	15.3
145	32.947	20
146	20.85\n42.394	46.5\n2.2
147	40.471	15.4
148	35.715	18.5

149 rows × 2 columns

```
print(df.shape)
```

(149, 5)

```
print(df_categorical_data.shape)
```

(149, 3)

```
print(df_numerical_data.shape)
```

(149, 2)

```
print(df.columns)
```

Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
 'IncomeGroup'],
 dtype='object')

```
print(df_numerical_data.columns)
```

Index(['BirthRate', 'InternetUsers'], dtype='object')

```
df[:]
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola\nAlbania	AGO\nALB	45.985\n12.877	19.1\n57.2	Upper middle income\nUpper middle income
3	United Arab Emirates	ARE	11.044	88	High income
4	Argentina	ARG	17.716	59.9	High income
...	...	...	...	...	...
144	Samoa	WSM	26.172	15.3	Lower middle income
145	Yemen, Rep.	YEM	32.947	20	Lower middle income
146	South Africa\nCongo, Dem. Rep.	ZAF\nCOD	20.85\n42.394	46.5\n2.2	Upper middle income\nLow income
147	Zambia	ZMB	40.471	15.4	Lower middle income
148	Zimbabwe	ZWE	35.715	18.5	Low income

149 rows × 5 columns

df[10:11]

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
10	Benin	BEN	36.44	4.9	Low income

df[0:100:20]

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
20	Brunei Darussalam\nBhutan	BRN\nBTN	16.405\n18.134	64.5\n29.9	High income\nLower middle income
40	Eritrea	ERI	34.8	0.9	Low income
60	Indonesia	IDN	20.297	14.94	Lower middle income
80	Lesotho\nLithuania	LSO\nLTU	28.738\n10.1	5\n68.4529	Lower middle income\nHigh income

df[:, :5]

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
5	Armenia\nAntigua and Barbuda	ARM\nATG	13.308\n16.447	41.9\n63.4	Lower middle income\nHigh income
10	Benin	BEN	36.44	4.9	Low income
15	Belarus	BLR	12.5	54.17	Upper middle income
20	Brunei Darussalam\nBhutan	BRN\nBTN	16.405\n18.134	64.5\n29.9	High income\nLower middle income
25	China	CHN	12.1	45.8	Upper middle income
30	Costa Rica	CRI	15.022	45.96	Upper middle income
35	Djibouti\nDenmark	DJI\nDNK	25.486\n10	9.5\n94.6297	Lower middle income\nHigh income
40	Eritrea	ERI	34.8	0.9	Low income
45	Micronesia, Fed. Sts.	FSM	23.511	27.8	Lower middle income
50	Gambia, The\nGuinea-Bissau	GMB\nGNB	42.525\n37.503	14\n3.1	Low income\nLow income
55	Guam	GUM	17.389	65.4	High income
60	Indonesia	IDN	20.297	14.94	Lower middle income
65	Israel\nItaly	ISR\nITA	21.3\n8.5	70.8\n58.4593	High income\nHigh income
70	Kyrgyz Republic	KGZ	27.2	23	Lower middle income
75	Liberia	LBR	35.521	3.2	Low income
80	Lesotho\nLithuania	LSO\nLTU	28.738\n10.1	5\n68.4529	Lower middle income\nHigh income
85	Madagascar	MDG	34.686	3	Low income
90	Montenegro	MNE	11.616	60.31	Upper middle income
95	Malaysia\nNamibia	MYS\nNAM	16.805\n29.937	66.97\n13.9	Upper middle income\nUpper middle income
100	Norway	NOR	11.6	95.0534	High income
105	Philippines	PHL	23.79	37	Lower middle income
110	French Polynesia\nQatar	PYF\nQAT	16.393\n11.94	56.8\n85.3	High income\nHigh income
115	Senegal	SEN	38.533	13.1	Lower middle income
120	Serbia	SRB	9.2	51.5	Upper middle income
125	Sweden\nSwaziland	SWE\nSWZ	11.8\n30.093	94.7836\n24.7	High income\nLower middle income
130	Tajikistan	TJK	30.792	16	Lower middle income
135	Tanzania	TZA	39.518	4.4	Low income
140	St. Vincent and the Grenadines\nVenezuela, RB	VCT\nVEN	16.306\n19.842	52\n54.9	Upper middle income\nHigh income
145	Yemen, Rep.	YEM	32.947	20	Lower middle income

df[: :-1]

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
148	Zimbabwe	ZWE	35.715	18.5	Low income
147	Zambia	ZMB	40.471	15.4	Lower middle income
146	South Africa\nCongo, Dem. Rep.	ZAF\nCOD	20.85\n42.394	46.5\n2.2	Upper middle income\nLow income
145	Yemen, Rep.	YEM	32.947	20	Lower middle income
144	Samoa	WSM	26.172	15.3	Lower middle income
...	...	...	...	...	...
4	Argentina	ARG	17.716	59.9	High income
3	United Arab Emirates	ARE	11.044	88	High income
2	Angola\nAlbania	AGO\nALB	45.985\n12.877	19.1\n57.2	Upper middle income\nUpper middle income
1	Afghanistan	AFG	35.253	5.9	Low income
0	Aruba	ABW	10.244	78.9	High income

149 rows × 5 columns

df[::-5]

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
148	Zimbabwe	ZWE	35.715	18.5	Low income
143	Vanuatu\nWest Bank and Gaza	VUT\nPSE	26.739\n30.394	11.3\n46.6	Lower middle income\nLower middle income
138	United States	USA	12.5	84.2	High income
133	Trinidad and Tobago	TTO	14.59	63.8	High income
128	Chad\nTogo	TCD\nTGO	45.745\n36.08	2.3\n4.5	Low income\nLow income
123	Slovak Republic	SVK	10.1	77.8826	High income
118	Sierra Leone	SLE	36.729	1.7	Low income
113	Rwanda\nSaudi Arabia	RWA\nSAU	32.689\n20.576	9\n60.5	Low income\nHigh income
108	Portugal	PRT	7.9	62.0956	High income
103	Pakistan	PAK	29.582	10.9	Lower middle income
98	Nigeria\nNicaragua	NGA\nNIC	40.045\n20.788	38\n15.5	Lower middle income\nLower middle income
93	Mauritius	MUS	10.9	39	Upper middle income
88	Mali	MLI	44.138	3.5	Low income
83	Macao SAR, China\nMorocco	MAC\nMAR	11.256\n21.023	65.8\n56	High income\nLower middle income
78	Liechtenstein	LIE	9.2	93.8	High income
73	Kuwait	KWT	20.575	75.46	High income
68	Japan\nKazakhstan	JPN\nKAZ	8.2\n22.73	89.71\n54	High income\nUpper middle income
63	Iraq	IRQ	31.093	9.2	Upper middle income
58	Croatia	HRV	9.4	66.7476	High income
53	Grenada\nGreenland	GRD\nGRL	19.334\n14.5	35\n65.8	Upper middle income\nHigh income
48	Ghana	GHA	33.131	12.3	Lower middle income
43	Finland	FIN	10.7	91.5144	High income
38	Ecuador	ECU	21.07	40.353684	Upper middle income
33	Czech Republic	CZE	10.2	74.1104	High income
28	Colombia	COL	16.076	51.7	Upper middle income
23	Canada\nSwitzerland	CAN\nCHE	10.9\n10.2	85.8\n86.34	High income\nHigh income
18	Brazil	BRA	14.931	51.04	Upper middle income
13	Bahrain	BHR	15.04	90.00004	High income
8	Azerbaijan\nBurundi	AZE\nBDI	18.3\n44.151	58.7\n1.3	Upper middle income\nLow income
3	United Arab Emirates	ARE	11.044	88	High income

df.isnull()

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
144	False	False	False	False	False
145	False	False	False	False	False
146	False	False	False	False	False
147	False	False	False	False	False
148	False	False	False	False	False

149 rows × 5 columns

df.isnull().sum()

	0
CountryName	0
CountryCode	0
BirthRate	0
InternetUsers	0
IncomeGroup	0

dtype: int64

df.isna()

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
144	False	False	False	False	False
145	False	False	False	False	False
146	False	False	False	False	False
147	False	False	False	False	False
148	False	False	False	False	False

149 rows × 5 columns

df.describe()

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
count	149	149	149.0	149.0	149
unique	149	149	139.0	143.0	18
top	Aruba	ABW	10.2	9.2	High income
freq	1	1	4.0	2.0	35

df.columns=['a','b','c','d','e']

df

	a	b	c	d	e
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola\nAlbania	AGO\nALB	45.985\n12.877	19.1\n57.2	Upper middle income\nUpper middle income
3	United Arab Emirates	ARE	11.044	88	High income
4	Argentina	ARG	17.716	59.9	High income
...	...	...	...	...	...
144	Samoa	WSM	26.172	15.3	Lower middle income
145	Yemen, Rep.	YEM	32.947	20	Lower middle income
146	South Africa\nCongo, Dem. Rep.	ZAF\nCOD	20.85\n42.394	46.5\n2.2	Upper middle income\nLow income
147	Zambia	ZMB	40.471	15.4	Lower middle income
148	Zimbabwe	ZWE	35.715	18.5	Low income

149 rows × 5 columns

```
df.columns = ['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers', 'IncomeGroup']
df
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola\nAlbania	AGO\nALB	45.985\n12.877	19.1\n57.2	Upper middle income\nUpper middle income
3	United Arab Emirates	ARE	11.044	88	High income
4	Argentina	ARG	17.716	59.9	High income
...	...	...	...	...	...
144	Samoa	WSM	26.172	15.3	Lower middle income
145	Yemen, Rep.	YEM	32.947	20	Lower middle income
146	South Africa\nCongo, Dem. Rep.	ZAF\nCOD	20.85\n42.394	46.5\n2.2	Upper middle income\nLow income
147	Zambia	ZMB	40.471	15.4	Lower middle income
148	Zimbabwe	ZWE	35.715	18.5	Low income

149 rows × 5 columns

```
df['BirthRate'] = df['BirthRate'].astype(str).apply(lambda x: x.split('\n')[0])
df['InternetUsers'] = df['InternetUsers'].astype(str).apply(lambda x: x.split('\n')[0])

df['BirthRate'] = pd.to_numeric(df['BirthRate'], errors='coerce')
df['InternetUsers'] = pd.to_numeric(df['InternetUsers'], errors='coerce')

df.BirthRate * df.InternetUsers
```

```
0
0    808.2516
1    207.9927
2    878.3135
3    971.8720
4   1061.1884
...
144   400.4316
145   658.9400
146   969.5250
147   623.2534
148   660.7275
```

149 rows × 1 columns

dtype: float64

```
df['mycalc']=df.BirthRate*df.InternetUsers
```

df

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	mycalc
0	Aruba	ABW	10.244	78.9	High income	808.2516
1	Afghanistan	AFG	35.253	5.9	Low income	207.9927
2	Angola\nAlbania	AGO\nALB	45.985	19.1	Upper middle income\nUpper middle income	878.3135
3	United Arab Emirates	ARE	11.044	88.0	High income	971.8720
4	Argentina	ARG	17.716	59.9	High income	1061.1884
...	...	...	...	...	...	...
144	Samoa	WSM	26.172	15.3	Lower middle income	400.4316
145	Yemen, Rep.	YEM	32.947	20.0	Lower middle income	658.9400
146	South Africa\nCongo, Dem. Rep.	ZAF\nCOD	20.850	46.5	Upper middle income\nLow income	969.5250
147	Zambia	ZMB	40.471	15.4	Lower middle income	623.2534
148	Zimbabwe	ZWE	35.715	18.5	Low income	660.7275

149 rows × 6 columns

```
df=df.drop('mycalc',axis=1)
```

df

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola\nAlbania	AGO\nALB	45.985	19.1	Upper middle income\nUpper middle income
3	United Arab Emirates	ARE	11.044	88.0	High income
4	Argentina	ARG	17.716	59.9	High income
...	...	...	...	...	...
144	Samoa	WSM	26.172	15.3	Lower middle income
145	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
146	South Africa\nCongo, Dem. Rep.	ZAF\nCOD	20.850	46.5	Upper middle income\nLow income
147	Zambia	ZMB	40.471	15.4	Lower middle income
148	Zimbabwe	ZWE	35.715	18.5	Low income

149 rows × 5 columns

df['InternetUsers']

	InternetUsers
0	78.9
1	5.9
2	19.1
3	88.0
4	59.9
...	...
144	15.3
145	20.0
146	46.5
147	15.4
148	18.5

149 rows × 1 columns

dtype: float64

df['InternetUsers']<2

	InternetUsers
0	False
1	False
2	False
3	False
4	False
...	...
144	False
145	False
146	False
147	False
148	False

149 rows × 1 columns

dtype: bool

```
df[df['InternetUsers']<2]
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
40	Eritrea	ERI	34.800	0.9	Low income
42	Ethiopia	ETH	32.925	1.9	Low income
49	Guinea	GIN	37.337	1.6	Low income
97	Niger	NER	49.661	1.7	Low income
118	Sierra Leone	SLE	36.729	1.7	Low income

```
len(df[df['InternetUsers']<2])
```

5

```
df[df['BirthRate']>40]
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
2	Angola\nAlbania	AGO\nALB	45.985	19.1	Upper middle income\nUpper middle income
11	Burkina Faso\nBangladesh	BFA\nBGD	40.551	9.1	Low income\nLower middle income
50	Gambia, The\nGuinea-Bissau	GMB\nGNB	42.525	14.0	Low income\nLow income
88	Mali	MLI	44.138	3.5	Low income
97	Niger	NER	49.661	1.7	Low income
98	Nigeria\nNicaragua	NGA\nNIC	40.045	38.0	Lower middle income\nLower middle income